

# ajedrezUPV - Project Documentation

Artificial Intelligence Chess Engine Project

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# 1 Project Overview

**ajedrezUPV** is an AI chess project from UPV. Focus: decision algorithms, heuristic evaluation, game engine programming.

## Key Features

- Custom chess engine (C++23)
- Alpha-Beta pruning algorithm
- UCI notation support
- Neural network integration
- Modular and extensible code
- Human vs AI gameplay

## 2 Repository Structure

Reorganized with English names:

| Path                      | Description                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------|
| .<br>README.md<br>LICENSE | Root<br>Main documentation<br>CC BY-NC 4.0                                                                   |
| docs/                     | Documentation<br>FEN.pdf<br>PROJECT_DOCUMENTATION.tex<br>PROJECT_DOCUMENTATION.pdf                           |
| data/                     | Data files<br>dataset.csv<br>mnist_loader.py                                                                 |
| models/                   | Models<br>modelo.py                                                                                          |
| tests/                    | Tests<br>prova1.py                                                                                           |
| implementations/          | C++ engine<br>alphabetaBot.cpp<br>botConMinimaxOptimizado.cpp<br>NodeMove.cpp/h<br>redNeuronalSquareRook.cpp |
| traditional_evaluation/   | Evaluators<br>EndgameEvaluator.cpp/h<br>Evaluator.cpp/h<br>GeneralEvaluator.cpp/h<br>OpeningEvaluator.cpp/h  |
| neural_network/           | Neural Network<br>jsonExtractor.cpp<br>json.hpp<br>modificacionesRedNeuronal.cpp                             |
| neural_network_jr/        | NN Junior<br>jsonExtractorJR.cpp<br>redNeuronalJR.cpp                                                        |
| lichess_db/               | Lichess DB<br>chunk_000 to chunk_019.jsonl                                                                   |

## 3 Chess Engine

C++ implementation with Alpha-Beta algorithm using Disservin library.

### Key Components

- `alphabetaBot.cpp` - Main game loop, UCI input, console display
- `NodeMove.cpp/h` - Game tree node with `alphaBeta()` method
- `GeneralEvaluator.cpp` - Position evaluation
- `OpeningEvaluator.cpp` - Opening heuristics
- `EndgameEvaluator.cpp` - Endgame heuristics

## 4 Connect Four AI

Neural network AI for Connect Four. Moved to its own repository: **connectUPV** (<https://github.com/LLPB-pixel/connectUPV>).

## 5 Neural Networks

- `neural_network/` - Main NN with `jsonExtractor`
- `neural_network_jr/` - Junior version

## 6 Technologies

C++: Disservin, nlohmann/json, STL. Python: TensorFlow, NumPy, scikit-learn, Jupyter.

## 7 Installation

Compile: `g++ implementations/*.cpp traditional_evaluation/*.cpp -I. -o chess_engine`  
Run: `./chess_engine` (UCI notation: e2e4, etc.)  
Connect Four is no longer part of this repository (see **connectUPV**).

## 8 License

CC BY-NC 4.0 - <https://creativecommons.org/licenses/by-nc/4.0/>  
Allowed: Share, Adapt, Educational/Personal/Research use.  
Not Allowed: Commercial use without permission.

## 9 Status

Completed: Chess engine, Alpha-Beta, evaluation system, victory check.  
In Progress: Advanced NN for chess, move explorer, multi-threading, opening book.

## 10 Contributing

Llorenc Perez - [perezllorenc@gmail.com](mailto:perezllorenc@gmail.com) - <https://github.com/LLPB-pixel/ajedrezUPV>

## 11 Acknowledgments

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<https://github.com/LLPB-pixel/ajedrezUPV>